Written examination date: May 24th, 2023
Course title: Introduction to reinforcement learning and control
Course number: 02465
Aids allowed: All aids allowed
Exam duration: 4 hours

Weighting: The exam is divided into 3 parts:

- Multiple-Choice questions
- Conceptual questions
- Programming questions

The overall scores of each part contribute roughly equally towards the overall result. Each question in each part contribute equally towards the score of that part.

Part I: Questions 1-9 are multiple choice. The score of a correct answer is 3 points. The score of an incorrect answer is -1 points. The score of option E or blank is 0 points.

Part II and part III: Each completed sub-task contribute towards your score.

Preparing your handin: The three parts are prepared as follows:

Part I: Edit the file `irlc/exam/exam2023spring/multiple_choice_answers.py`. Don't include calculations. Only answer with `'A'`, `'B'`, `'C'`, `'D'`, `'E'`.

Part II: Create a PDF file with your answers and justifications.

Part III: Program your answer in the `.py`-files indicated in the question and run `irlc/exam/exam2023spring/exam2023spring_tests_grade.py` to generate your `.token`-file.

Handing in: To hand in, you should upload the files:

- The `irlc/exam/exam2023spring/multiple_choice_answers.py`-file with your answer to part I
- The `.pdf`-file with your answers to part II
- The `.token`-file with your answers to part III
- The `irlc/exam/exam2023spring/question_inventory.py` source file containing your solution
- The `irlc/exam/exam2023spring/question_lqr.py` source file containing your solution
- The `irlc/exam/exam2023spring/question_bandit.py` source file containing your solution

Note on part II: The main quantities asked for are highlighted as $f(x)$. Answer unambigiously, concisely, and if applicable with algebraic simplifications. Your final result must be clearly indicated at the end of your answer: $f(x) = 3\sin(x)$. To get credit, you must state the relevant theory and equations, and all relevant calculations must be included. Credit is not given for answers with missing or erroneous justifications.

Note on part III: To get started, move the folder `irlc/exam/exam2023spring` from the `.zip` file to the corresponding location in your existing exercise directory. The `.py` source files must be **reproducible** and **readable** so that someone else can run and fully understand your solution. You can freely use the `irlc`-toolbox (including solutions) and the packages we have used in the course, but you **must** include additional code you write during the exam or have prepared beforehand in the source files listed above. The source files must not be renamed. The `.token` file contains your results and must be up to date with your source files, i.e., generate it just prior to handin. In the case they differ, the `.token` file takes precedence. Credit is given for correct implementations defined by the problem description. The points in the `.token` file name is computed from the public tests, and might not correspond to overall correctness.

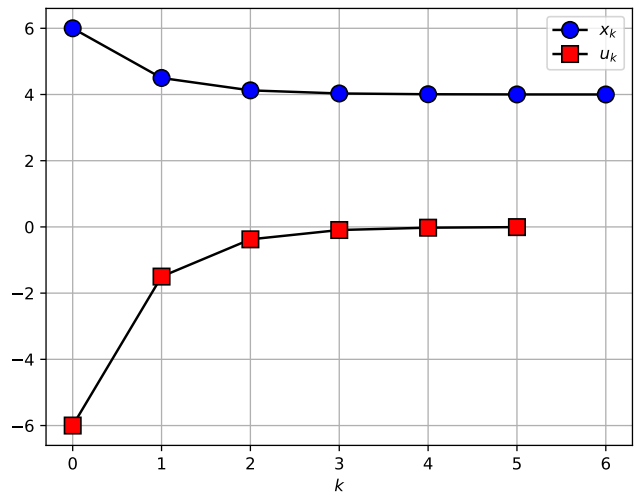


Figure 1: Plot of PID controller

Part I: Multiple-Choice

Question 1: .....

Suppose we want to apply the dynamical programming algorithm to chess [1]. This will lead to several practical problems, however, focusing just on the potential problems listed below, which one will be a main obstacle in terms of obtaining a near-perfect chess policy using the dynamical programming algorithm?

A. Within a few iterations, the policy function μ_k will require too much memory to store

B. Given a state $\boldsymbol{x}_k$, it is not practical to define the action spaces $\mathcal{A}_k(x_k)$

C. It will require too much space to store the state space $\mathcal{S}_2$

D. There is no reasonable choice of planning horizon N

E. Don't know.

Solution: The correct answer is A.

A. It will be impossible to store the policy as it requires as much space as the size of the state space. I.e. policy μ_{N-1} (the first we would potentially compute) would require $|\mathcal{S}_{N-1}|$ memory.

B. The action spaces are just the available actions, and can thus be derived from the rules of chess (each action space contains about 10 elements)

C. The second state space is still reasonably small, containing about 10^4 states

D. Considering a typical game is about 45 moves, a planning horizon of $N = 100$ would easily be sufficient

Question 2: .....

Consider PID control applied to steer a car along a straight track. The control signal u_k corresponds to the angle between the front wheel and the centerline of the track, the input signal x_k corresponds to the angle between the car body and the track in degrees, and the goal of the PID controller is to bring the angle between the car body and the track to a value of $x^* = 4$ degrees (corresponding to executing

[1] We don't consider time control as part of the problem.

Question 2 continues on the next page...

a turn). Figure 1 shows the behavior of both x_k and u_k at time steps $k = 0, 1, 2, \ldots$. Suppose the PID controller takes the form described in the lecture notes, and assume $K_d = K_i = 0$, which one of the following options are true?

A. $K_p = 1$

B. $K_p = 2$

C. $K_p = 3$

D. There is not enough information to determine the correct answer

E. Don't know.

Solution: The correct answer is C.

The initial error is $e = x^* - x = 4 - 6.0 = -2.0$. We can therefore compute

$$K_p = \frac{u_0}{e} = \frac{-6.0}{-2.0} = 3.$$

	Arm $a = 0$	Arm $a = 1$	Arm $a = 2$	Arm $a = 3$
$N_t(a)$	14	11	6	3
Total reward $S_t(a)$	45	31	15	5

Table 1: Outcome of simulating a k-armed bandit problem for a few iterations

Question 3: .....

Which one of the following options are correct about direct control using trapezoid collocation?

A. It is an example of open-loop control

B. It cannot take constraints into account

C. It requires us to specify a fixed initial state $\boldsymbol{x}_0 = \boldsymbol{x}(t_0)$

D. It can only be applied to quadratic cost functions

E. Don't know.

Solution: The correct answer is A.

A. Direct control computes a single control path $\boldsymbol{u}(t)$ and is therefore an example of open-loop control

B. Direct control takes constraints into account as part of the optimization; c.f. the pendulum swingup example.

C. Neither the initial or terminal state needs to be fixed.

D. Although we only consider quadratic cost functions in the exercises, direct control is formulated for a general cost function

Question 4: .....

Suppose iterative LQR is applied to steer the racecar from the car-example ([Her25, section 10.4.3]) through the track. Which of the following statements are true about the dimensions of the linearized matrices A_k and B_k?

A. The dimension of A_k is 6×6 and the dimension of B_k is 6×2

B. The dimension of A_k is 6×2 and the dimension of B_k is 6×1

C. The dimension of A_k is 2×2 and the dimension of B_k is 2×6

D. The dimension of A_k is 6×6 and the dimension of B_k is 2×2

E. Don't know.

Solution: The correct answer is A.

The state has 6 dimensions and the control signal has 2 dimensions. It follows that A_k must be 6×6 and the dimension of B_k must be 6×2, otherwise it could not be the case that

$$\boldsymbol{x}_{k+1} = A_k \boldsymbol{x}_k + B_k \boldsymbol{u}_k + \boldsymbol{d}_k$$

Question 5: .....

Consider a k-armed bandit problem with $k = 4$. Suppose we are at time step t, and we denote by $N_t(a)$ the total number of times we have pulled arm $a = 0, 1, 2, 3$ up to now, and by $S_t(a)$ the sum of all

Question 5 continues on the next page...

rewards we have obtained by pulling arm a, i.e. in the notation [SB18][Eq. 2.1]

$$S_t(a) = \{\text{Sum of rewards when } a \text{ taken prior to } t\} = \sum_{i=1}^{t-1} R_i \cdot \mathbb{1}_{A_i=a}$$

$$N_t(a) = \{\text{Number of times } a \text{ taken prior to } t\} = \sum_{i=1}^{t-1} \mathbb{1}_{A_i=a}.$$

These values have been collected in table 1. Suppose we apply UCB1 to the problem using $c = 2$. Which arm will be selected next?

A. UCB1 selects arm $A_t = 0$

B. UCB1 selects arm $A_t = 1$

C. UCB1 selects arm $A_t = 2$

D. UCB1 selects arm $A_t = 3$

E. Don't know.

Solution: The correct answer is A.

To solve this problem, recal the definition of the UCB1 algorithm that it selects the arm according to

$$A_t = \arg\max_a \left[\frac{S_t(a)}{N_t(a)} + c\sqrt{\frac{\log(t)}{N_t(a)}} \right]$$

where $t = 1 + \sum_{a=0}^{k-1} N_t(a)$. This gives the three upper-confidence values:

$$q_0 = 4.222, \quad q_1 = 3.955, \quad q_2 = 4.04, \quad q_3 = 3.844.$$

Using this, it is easy to see that arm $a = 0$ will be selected as it has the higher upper-confidence bound

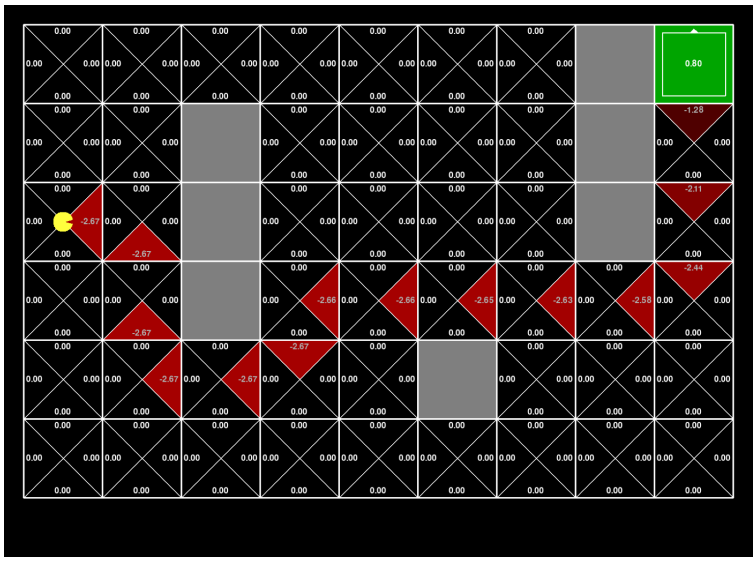


Figure 2: A gridworld environment and the Q-values updated by applying first-visit Monte-Carlo for one episode. The plot is for illustration purposes, and the problem can be solved without consulting the particular values shown here.

Question 6: .....

Let v_*, q_* be the optimal value and action-value functions of an MDP, let π be any policy and finally let v_π and q_π be the value/action-value function associated with π. Which one of the following statements are true in general?

A. $\max_s q_*(s,a) = v_*(a)$

B. There is a policy π, a state s and an action a so that $q_*(s,a) < q_\pi(s,a)$

C. For all π and a it is true that $q_*(s,a) > q_\pi(s,a)$

D. There is a policy π and state s so that $\max_a q_*(s,a) = v_\pi(s)$

E. Don't know.

Solution: The correct answer is D.

A. This statement is meaningless since v_* is defined on states not actions

B. This is false since q_* is obtained by maximizing over policies

C. This is false when π is the optimal policy

D. This is true for all states when π is the optimal policy

Question 7: .....

We consider the familiar gridworld environment discussed in [Her25, section 4.2.4] shown in fig. 2. The agent receives a reward of $+1$ on completion (the upper-right square), and otherwise a living reward of -2. Recall that in first visit Monte-Carlo control (see [SB18, Section 5.3]), the action-values $Q(s,a)$ are computed as the average of the returns. Suppose that during a particular episode, the returns computed by first-visit monte carlo are $G_0, G_1, \ldots, G_{T-1}$ (for illustration purposes, fig. 2 shows a single update of first-visit monte carlo with our choice of rewards, and for particular choices of γ, α). What relationship holds true for the returns in the beginning of the episode, i.e. $t \leq T-2$?

A. $G_t = \gamma G_{t+1} - 2$

Question 7 continues on the next page...

B. $G_{t+1} = \alpha G_t - 2\gamma + G_T \gamma^T$

C. $G_t = \alpha G_{t+1} \gamma^t$

D. $G_t = -2\gamma^t + 1$

E. Don't know.

Solution: The correct answer is A.

This problem can be solved by observing how the return are iteratively computed in the first-visit Monte-Carlo method or by inspecting the definition. In either case it is true that

$$G_t = R_{t+1} + \gamma G_{t+1}.$$

Question 8: .....

Which of the following statements are true about TD(λ)?

A. TD(λ) cannot be used with function approximators

B. The role of the eligibility trace is to let reward obtained earlier in an episode affect the change in the value function later in the episode

C. The eligibility trace cannot be negative

D. The eligibility trace is a measure of the amount of reward obtained in a given state weighted by an exponential factor

E. Don't know.

Solution: The correct answer is C.

A. $TD(\lambda)$ works excellently with function approximators (one of the early successes in reinforcement learning, TD-gammon in the mid 90s, used $TD(\lambda)$ with function approximators)

B. The role is to interpolate between Monte-Carlo learning and $TD(0)$. The eligibility trace allows reward obtained later in the episode to propagate backwards in time.

C. The eligibility trace cannot be negative by definition (it is defined as sum / products of positive numbers)

D. The reward function has no influence on the eligibility trace

Question 9: .....

Which one of the following statements about Q-learning is correct?

A. The first step in training a Q-learning agent is to compute the set of all states the agent can be in

B. The Q-table $Q(s, a)$ in Q-learning is a measure of the reward the agent will obtain in the very next step multiplied by γ

C. Q-learning still works if we initialize the Q-table to -1, i.e. $Q(s, a) = -1$ for all $s \in \mathcal{S}$

D. When Q-learning is applied to a deterministic environment, the agent will follow a deterministic policy

E. Don't know.

Solution: The correct answer is C.

A. Q-learning does not require us to know the set of all reachable states from the onset (although if it is run to convergence it will end up computing them)

B. The Q-table measures the discounted *accumulated* reward

C. This is true. Q-learning forgets the initial value of the Q-table (although it may affect exploration depending on the reward function)

D. This is not true. Q-learning requires exploration to work, hence it will sometimes act randomly

Part II: Conceptual questions

Question 10: .....

Consider a control problem where a control signal $u(t) \in \mathbb{R}$ is applied to control a system with state $x(t) \in \mathbb{R}$, and where the dynamics satisfy the following differential equation:

$$\dot{x} = f(x, u) = 4ux \tag{1}$$

The first two questions will assume the problem has been discretized using a time constant of $\Delta = 0.5$ to yield states $x_0, x_1, x_2, \cdots$ and control signals $u_0, u_1, u_2, \ldots$.

(a) Assume the problem is Euler discretized. Determine the discrete dynamics f_k used to compute $x_{k+1} = f_k(x_k, u_k)$.

Solution: Simply inserting into the equation for Euler discretization, $x_{k+1} = x_k + \Delta f(x_k, u_k)$, we get:

$$f_k(x_k, u_k) = 2.0 u_k x_k + x_k \tag{2}$$

(b) Continuing the previous problem, suppose we wish to apply a LQR controller to control the system near a state $\bar{x}$. The system is therefore linearized around $\bar{x}$ and $\bar{u} = 1$ to give rise to the linearized dynamics $x_{k+1} = Ax_k + Bu_k + d$. Determine A, B and d in terms of $\bar{x}$.

Solution: The general linearized expression is:

$$f_k(x_k, u_k) \approx 2.0\bar{u}\bar{x} + 2.0\bar{x}\left(-\bar{u} + u_k\right) + \bar{x} + \left(2.0\bar{u} + 1\right)\left(-\bar{x} + x_k\right)$$

and inserting the known value for $\bar{u}$ we get:

$$f_k(x_k, u_k) \approx 2.0\bar{x}\left(u_k - 1\right) + 3.0x_k$$

So that $A = \begin{bmatrix}3.0\end{bmatrix}$, $B = \begin{bmatrix}2.0\bar{x}\end{bmatrix}$ and $d = \begin{bmatrix}-2.0\bar{x}\end{bmatrix}$.

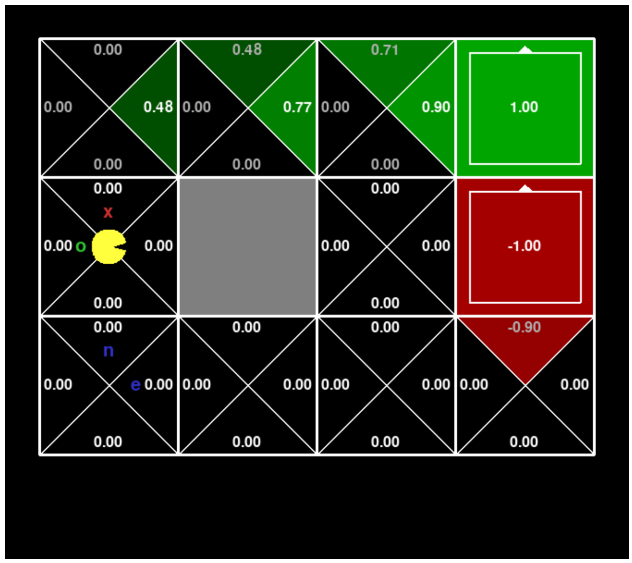


Figure 3: An example gridworld environment that Sarsa is applied to. The figure shows the current Q-values.

Question 11: .....

Suppose that Sarsa (using discount factor $\gamma = 1$, a learning rate $\alpha = 0.9$ and an exploration rate of $\varepsilon = 0.1$) is applied to the Gridworld MDP shown in fig. 3. The living reward is 0, and the dynamics is deterministic. Recall that pacman stay at the current state if he choose an action which moves him into a wall. The current state of the agent is as indicated in the figure.

(a) Suppose that in the next step of the Sarsa algorithm, the agent takes (and execute) the action `North` in the current position s. Upon taking this action, the Q-value associated with the red cross $Q(s, \texttt{North})$ will be updated by Sarsa.

What are the possible value(s) of $Q(s, \texttt{North})$ after this step? (if there are more than one, list all of them).

Solution: Since the immediate reward is zero, the next Q-value will be determined by the Q-value associated with the state north of the agent and the action the agent generates in that state:

$$Q(s, a) = Q(s, a) + \alpha(r + \gamma Q(s', a') - Q(s, a))$$

If the exploration rate is non-zero, all actions a' may occur, giving rise to two different new values. This mean the Q-value can be updated to:

$$Q(s, \texttt{North}) = 0.0, 0.432$$

(b) In the previous question, we took one step, performed a single action `North`, and updated one Q-value.

In this question, suppose again that the agent starts in the position indicated in fig. 3 and assume Sarsa is applied to update the Q-values shown in the figure.

Different sequences of future actions will result in different Q-values being updated. What are the minimum number of steps which are required before the Q-value associated with the green circle can take a value different than 0, and what actions will the agent take in this case? Give your answer as a list of actions.

Question 11 continues on the next page...

Solution: It is evident that we need to propagate the Q-value from the northern square to the Q-value we wish to update. To do this, we first go `north`, but then to change the Q-value we must select `east` in that state. Therefore, we backtrack (`west`, `south`). Pacman will now be on the current state, and a single step `west` will mean pacman updates the green Q-value, and it can be updated to a non-zero value since the next action generated by Sarsa can select the Q-value associated by the red square by moving $A' =$ `north`. The answer is therefore 5 and the actions are:

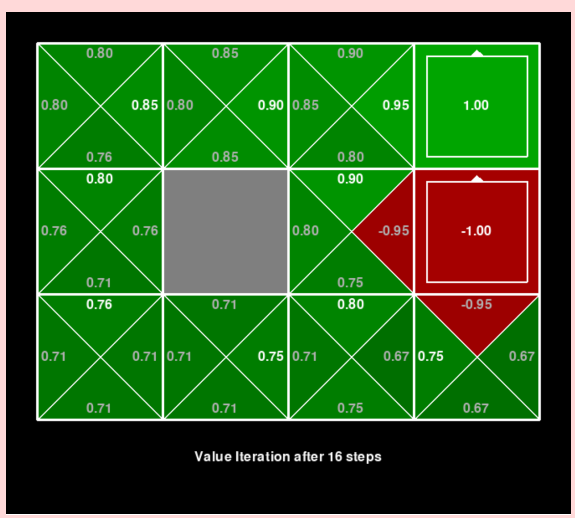
`north`, `east`, `west`, `south`, `west`

Note: Since the question asks for the number of steps this is the right answer. However, an answer that included the last (not yet executed) `north`-action would also have been accepted. If you want, you can re-create the example using the Sarsa-demo in `irlc/lectures/lec11/lecture_11_sarsa.py`

(c) Sarsa learning has clearly not converged in the example shown above. However, assume we apply a more realistic version of Sarse where $\gamma = .95$, and where importantly α decrease to 0 at a rate satisfying the stochastic approximation conditions [SB18, Eq. (2.7)] for convergence so that the Q-values converge to their *true values* under Sarsa.

Consider the two Q-value associated with moving `North` and `East` Q_n, Q_e, indicated by the blue n, e-letters in fig. 3. After convergence, it must be the case that either they will have the same value, $Q_n = Q_e$, or one will be greater than the other: $Q_n > Q_e$ or $Q_n < Q_e$. State which is the case and provide a clear and specific argument for your answer.

Solution: After convergence, Sarsa will have learned the Q-values associated with the ε-soft policy π, i.e. to q_π. It will clearly attempt to move the agent towards the goal square with a $+1$ reward, and since $\gamma < 1$ it will attempt to do so quickly. The fastest way to do that is either north or south of the central pillar. The southern way, associate with Q_e, takes the agent next to the dangerous square with a -1 reward. There is a chance of at least $\frac{\varepsilon}{4}$ of randomly falling into that square using the ε-soft policy. We can therefore conclude this path is far more dangerous, and this must be reflected in the Q-values. Hence, $Q_e < Q_n$. Note that this will not be true for Q-learning, where the two paths are the same. Note this argument is similar to the cliffwalking example we saw in the exercises and in [SB18]. It can also be confirmed by inspecting the converged value of Sarsa shown below:



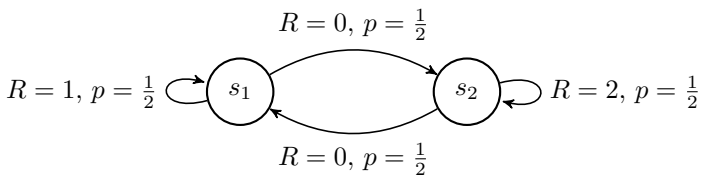


Figure 4: A simple MRP

Question 12: .....

Consider a Markov Reward Process with two states s_1 and s_2 and a discount factor of $0 < \gamma < 1$ shown in fig. 4 [2]. With equal probability $\frac{1}{2}$, the system will either stay in the current state or transition to the other state. The MRP never terminates.

- When the system transition s_1 to s_2 (or s_2 to s_1) it will receive a reward of 0,
- If it transition from s_1 to s_1 it will receive a reward of 1,
- If it transition from s_2 to s_2 it will receive a reward of 2.

Since there are two states, the value function v_π takes two values $v_1, v_2 \in \mathbb{R}$ defined as: $v_1 = v_\pi(s_1)$ and $v_2 = v_\pi(s_2)$.

(a) Assume for a moment that $v_2 = \frac{5}{2}$ and $\gamma = \frac{2}{3}$. What is v_1? (i.e. the value function in s_1, $v_\pi(s_1)$)

Solution: Recall Bellmans equation (it does not matter which) says: $v(s) = \mathbb{E}[R + \gamma v(s') \mid S_t = s]$. Therefore:

$$v_1 = \frac{1}{2}(R + \gamma v_1) + \frac{1}{2}\gamma v_2 = \frac{1}{2}(1 + \gamma v_1 + \gamma v_2)$$

Solving we get that $v_1 = \frac{1+\gamma v_2}{2-\gamma} = 2$

(b) Ignore the previous question and consider the general form of the problem. As the name suggest, when Bellmans expectation equations are applied to this problem we obtain two equations. Write them as a linear system of the form $\boldsymbol{b} = A\boldsymbol{v}$ where $\boldsymbol{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$. State what A and $\boldsymbol{b}$ are as functions of γ.

Solution: We already got the first equation from the previous problem. Both equations are:

$$-1 = (\gamma - 2)v_1 + \gamma v_2$$
$$-2 = (\gamma - 2)v_2 + \gamma v_1$$

So therefore

$$\begin{bmatrix} -1 \\ -2 \end{bmatrix} = \begin{bmatrix} \gamma - 2 & \gamma \\ \gamma & \gamma - 2 \end{bmatrix} \boldsymbol{v}$$

[2] Recall a Markov Reward Process is a Markov decision process without actions, see [SB18, Example 6.2].

Part III: Programming questions

Question 13: .....

To solve this question, you should edit the file `irlc/exam/exam2023spring/question_inventory.py`**.** This problem focuses on a variant of the inventory control problem discussed in [Her25, section 5.1.2]. This inventory problem represents a flower-store such that x_k denotes the number of flower bouquets in stock at planning round k. The original inventory control model and the dynamical programming algorithm is included in the exam folder.

The following tasks can be solved by implementing suitable variants of the inventory control problem, and then applying dynamical programming to determine the optimal policy $\mu_0^*(x_0)$ and cost-function $J^*(x_0)$ in the starting state.

The flower store problem is equivalent to the inventory control problem on a horizon of N with two changes:[3]:

- $g_k(x_k, u_k, w_k) = cu + |x_k + u_k - w_k|$
- The distribution of the number of items customers buy w_k is:

$$p_W(w_k = 0|x_k, u_k) = 0.1, \quad p_W(w_k = 1|x_k, u_k) = 0.3, \quad p_W(w_k = 2|x_k, u_k) = 0.6.$$

(a) Complete `def a_get_policy(N: int, c: float, x0 : int)`: This function is given a value of N, c and a starting state x_0, and should return the action the optimal policy computes in x_0, i.e. $\mu_0^*(x_0)$ as an `int`.

(b) Complete `def b_prob_one(N : int, x0 : int)`: For every policy and starting state x_0, there is a certain chance $p(x_N = 1|x_0)$ we will end up with a single item (bouquet) on the last day N when following the policy. The clerk operating the store would very much like to bring this last bouquet home with her, and so she is solely concerned with determining the policy which maximize $p(x_N = 1|x_0)$, i.e. the chance she can bring home a single bouquet at the end of the planning period.

Determine what this chance is when we follow the policy which is *solely* concerned with with maximizing the chance that $x_N = 1$. The function should accept N and x_0 as input argument, and return the value of $p(x_N = 1|x_0)$ as a `float`.

Hint: Alter the cost-functions so that the optimal solution maximize this probability. The Pacman-problems where we computed the probability of winning may provide inspiration.

Solution:

The solution can be found in this directory as a `.py` file.

[3]The expression $|x|$ return the absolute value, i.e. $|4| = |-4| = 4$

Question 14: .....

To solve this question, you should edit the file `irlc/exam/exam2023spring/question_lqr.py`**.** In this problem, we will consider the Pendulum-swingup task described in [Her25, section 10.4.1]. Recall that in this problem, the state of the pendulum is two-dimensional:

$$\boldsymbol{x} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

and under application of a control signal u, the state satisfies the differential equation:

$$\ddot{\theta} = \frac{g}{l}\sin(\theta) + \frac{u}{ml^2}, \quad g = 9.82,\, l = 1,\, m = 0.8. \tag{3}$$

(*The pendulum model is implemented as the class* `ContiniousPendulumModel` *in the file* `irlc/ex04/model_pendulum.py` *in the toolbox*). This model is assumed to be discretized using Euler discretization with a time constant of $\Delta = 0.5$ seconds, but *without* applying coordinate transformations. In other words, the discrete coordinates correspond to $\boldsymbol{x}_k = \begin{bmatrix} \theta_k \\ \dot{\theta}_k \end{bmatrix}$.

We want to eventually apply LQR to the problem. However, before we get to that we will consider how to simply apply discrete LQR to a comparable linear problem.

(a) Complete `def a_LQR_solve(a : float, x0 : np.ndarray)`: This function consider a discrete problem with cost matrices $Q = R = I$ and where the dynamics depends on a parameter a given by

$$\boldsymbol{x}_{k+1} = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \boldsymbol{x}_k + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_k + \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The function should apply discrete LQR to the problem on a horizon sufficiently long to guarantee convergence[4] of the policy matrices L_0, $\boldsymbol{l}_0$ and return the first action u_0 the discrete LQR controller wish to execute in state $\boldsymbol{x}_0$. The action should be returned as a `float`.

(b) Complete `def b_linearize(theta : float)`: Return to the general formulation eq. (3). Given an angle θ, this function should linearize the discretized model around $\bar{\boldsymbol{x}} = \begin{bmatrix} \theta \\ 0 \end{bmatrix}$ and $\bar{u} = 0$, and return three numpy `ndarray` objects A, B and $\boldsymbol{d}$ of the right dimension corresponding to the linearized problem. I.e., it should hold approximately for states and actions close to the linarization point $\bar{\boldsymbol{x}}$, $\bar{u}$ that

$$\boldsymbol{x}_{k+1} \approx A\boldsymbol{x}_k + Bu_k + \boldsymbol{d}$$

(c) Complete `def c_get_optimal_linear_policy(x0 : np.ndarray)`: We are now ready to build our controller. Given input state $\boldsymbol{x}_0$, specified as an `np.ndarray`, this function should linearize the pendulum problem around $\bar{\boldsymbol{x}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\bar{u} = 0$, then solve the linearized problem using LQR (using a sufficiently long horizon to guarantee convergence and using $Q = R = I$), and then finally return the action the LQR controller will take in the $\boldsymbol{x}_0$ as a `float`.

Solution:

The solution can be found in this directory as a `.py` file.

[4] For instance $N = 100$

Question 15: .....

To solve this question, you should edit the file `irlc/exam/exam2023spring/question_bandit.py`**.** In this question, you will use bandit algorithms to determine which of k actions, corresponding to advertisements, are the best based on the clicks-per-hour they result in when presented to users. The bandit algorithm we will first consider will be the simple bandit algorithm presented in [SB18][Section 2.4]. The data will be presented to the algorithm in the shape of a list of actions $(a_1, a_2, \ldots, a_{t-1})$ (`actions`) and a list of corresponding rewards $(r_1, r_2, \ldots, r_{t-1})$ (`rewards`). The actions will be assumed to be integers $0, 1, \ldots, k-1$.

(a) Complete `def a_select_next_action_epsilon0(k : int, actions : list, rewards : list)`: Given a number of arms k, a list of $t-1$ actions `actions` and rewards `rewards`, this function should return the next action a_t as an `int` generated according to the bandit algorithm described in [SB18][Section 2.4] when trained on the data, but assuming we act greedily all the time: $\varepsilon = 0$.

(b) Complete `def b_select_next_action(k : int, actions : list, rewards : list, epsilon : float)`: Consider again the simple bandit algorithm in [SB18][Section 2.4]. In addition to the previous input, the function should now also accept a exploration rate $\varepsilon \geq 0$. The function should return the action a_t as an `int` generated according to the simple bandit algorithm.

(c) Complete `def c_nonstationary_Qs(k : int, actions : list, rewards : list, alpha : float)`: This function should implement the non-stationary simple bandit algorithm with α-soft updates described in [SB18][Section 2.5]. The inputs have the same meaning as before, but the function should also accept the learning rate α as an input argument. The function should return a dictionary which has actions a as keys, and their corresponding Q-value $Q(a)$ as values.

Solution:

The solution can be found in this directory as a `.py` file.

Question 15 continues on the next page...

References

[Her25] Tue Herlau. Sequential decision making. (Freely available online), 2025.

[SB18] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. The MIT Press, second edition, 2018. (Freely available online).

This line concludes the exam. Document build: 2026/04/16, 22:35:14.